Boundary-free weak influences: range closure, sharp information and feasible GLS

1. Spaces and adjoint range

Let $J \in C^1([0,T];\mathbb{R}^{d\times d})$, and let the invertible fundamental matrix satisfy $Z' = JZ$, $Z(0) = I_d$. Put $Ah = h' - Jh$ and $A^*\phi = -\phi' - J^T\phi$. Sensor coordinates have independent stationary OU covariance with fixed positive scales. Its Cameron–Martin space $\mathcal{H}$ is H^1 with the OU inner product. Write $\mathcal{E} = \mathcal{H}'$ for its continuous dual, equipped with the covariance norm

$$\|\lambda\|_C^2 = \langle \lambda, C\lambda\rangle, \qquad \langle C^{-1}g, h\rangle = \langle g, h\rangle_{\mathcal{H}}.$$

This definition includes ordinary integrated influences and endpoint evaluation measures. It is the variance norm of the associated Gaussian linear observation; the noise path itself need not belong to $\mathcal{H}$.

Proposition 2 (exact range and closure). In L^2,

$$A^*H_0^1 = \left\{ \ell \in L^2 : \int_0^T Z(t)^T \ell(t)dt = 0 \right\}.$$

The range of compactly supported smooth tests is dense in this space in L^2. Its closure in $\mathcal{E}$ is exactly

$$\mathcal{M} = \overline{A^*C_c^\infty}^{\|\cdot\|_C} = \{\lambda \in \mathcal{E} : \langle \lambda, Z_j\rangle = 0, j = 1, \dots, d\}.$$

Consequently $\mathcal{M}$ has codimension d.

Proof. Since $Z^TA^*\phi = -(Z^T\phi)'$, zero endpoint values imply the moment constraint. Conversely, for an L^2 influence with zero moment,

$$\phi(t) = -Z(t)^{-T}\int_0^t Z(s)^T\ell(s)ds$$

belongs to H_0^1 and satisfies $A^*\phi = \ell$. Approximation in H_0^1 by C_c^∞ and continuity of $A^* : H_0^1 \to L^2$ give L^2 density. For the stronger closure statement, a function $h \in \mathcal{H}$ annihilates every $A^*\phi$ if and only if $Ah = 0$ distributionally, hence $h = Zc$. The Hahn–Banach annihilator characterization of a closed subspace then gives the stated closure in $\mathcal{H}'$. The columns of Z are linearly independent elements of $\mathcal{H}$, so their dual constraints have rank d. □

Zero values at the endpoints do not imply compact support. Equality for literal compact tests, without a closure or an H_0^1 qualification, would be false. The relevant closure uses the covariance norm, which is equivalent to the dual Sobolev norm for this positive OU model.

2. Sharp covariance reference

Let $S = [S_1, \dots, S_p] \in \mathcal{H}^p$ be coefficient sensitivities, and form the joint information blocks for S and Z. Assume $\mathcal{I}_{uu}$ and the Schur complement are positive definite. Set

$$R = S - Z\mathcal{I}_{uu}^{-1}\mathcal{I}_{u\theta}, \qquad \mathcal{I}_e = \mathcal{I}_{\theta\theta} - \mathcal{I}_{\theta u}\mathcal{I}_{uu}^{-1}\mathcal{I}_{u\theta}.$$

Projection and best linear unbiased estimation in correlated continuous regression are classical [1]. Here the adjoint-range characterization identifies exactly which dynamical weak influences obey that projection.

Theorem 2 (sharp boundary-free influence covariance). Let $\Lambda = [\lambda_1, \dots, \lambda_p] \in \mathcal{M}^p$ satisfy $\langle \lambda_r, S_s \rangle = \delta_{rs}$. Its Gaussian influence covariance obeys

$$V(\Lambda) \succeq \mathcal{I}_e^{-1}.$$

The bound is attained in $\mathcal{M}$ by the matrix-valued dual influence

$$\Lambda_* = C^{-1} R\, \mathcal{I}_e^{-1}.$$

It is the infimum for normalized compact-test influences. It is a minimum among literal compact tests only when Λ_* belongs to their adjoint range.

Proof. The $\mathcal{H}$ projection makes $\langle C^{-1}R, Z \rangle = 0$ and $\langle C^{-1}R, S \rangle = \mathcal{I}_e$. Thus Λ_* meets both constraints and has covariance $\mathcal{I}_e^{-1}$. The difference $\Delta = \Lambda - \Lambda_*$ pairs to zero with both S and Z, hence with R. Covariance cross terms vanish and

$$V(\Lambda) = \mathcal{I}_e^{-1} + V(\Delta).$$

For attainability as an infimum, approximate each column of Λ_* by compact-test influences in covariance norm. Their normalization matrices converge to I_p by continuous pairing with S; right-multiplication by their inverse transpose enforces exact normalization while preserving the range and covariance convergence. □

This is a matrix statement. Dividing a vector influence by a scalar information is valid only when $p = 1$. The information loss is exactly $\mathcal{I}_{\theta u}\mathcal{I}_{uu}^{-1}\mathcal{I}_{u\theta}$; covariance inflation is $\mathcal{I}_e^{-1} - \mathcal{I}_{\theta\theta}^{-1}$. There is no direction-independent scalar loss percentage in general.

3. Explicit OU representative

For a scalar coordinate with $g \in H^2$, integration by parts in the stationary OU norm gives the dual measure

$$C^{-1}g = \frac{g - \tau^2 g''}{2\sigma^2\tau}\, dt + \frac{g(0) - \tau g'(0)}{2\sigma^2}\delta_0 + \frac{g(T) + \tau g'(T)}{2\sigma^2}\delta_T.$$

For independent coordinates the formula applies coordinatewise, using their respective scales. If only $g \in H^1$, the Riesz representative remains well-defined in $\mathcal{H}'$ but need not be a density plus finite endpoint measures.

For a smooth optimal representative write $\Lambda_* = m(t)dt + m_0\delta_0 + m_T\delta_T$. On the open interval an adjoint test matrix is

$$\Phi_*(t) = -Z(t)^{-T}\left[Z(0)^T m_0 + \int_0^t Z(s)^T m(s)ds\right].$$

Extend it by zero outside the observation interval. Its endpoint jumps supply $m_0 = -\Phi_*(0+)$ and $m_T = \Phi_*(T-)$, and distributionally $A^*\Phi_* = \Lambda_*$. These formulas include the initial atom; omitting it in the cumulative integral is incorrect. The annihilation condition supplies the terminal jump. Cutoff and smoothing approximate this generalized test in covariance norm. These measures describe a statistical reference; the implemented estimators keep compact tests and add no explicit boundary observation terms.

For scalar exponential dynamics, $z = e^{at}$, $s = x_0te^{at}$, $k = \mathcal{J}_{au}/\mathcal{J}_{uu}$ and $g = e^{at}(x_0t - k)$. Define

$$A_0 = \frac{(1-\tau^2a^2)x_0}{2\sigma^2\tau}, \qquad B_0 = \frac{-(1-\tau^2a^2)k - 2\tau^2ax_0}{2\sigma^2\tau},$$

$$q_0 = \frac{-k - \tau(x_0 - ak)}{2\sigma^2}, \qquad F_j(t) = \int_0^t s^j e^{2as}ds.$$

Then

$$\phi_*(t) = -\frac{e^{-at}}{\mathcal{J}_e}[q_0 + A_0F_1(t) + B_0F_0(t)].$$

For $a \neq 0$, $F_0 = \mathrm{expm1}(2at)/(2a)$ and $F_1 = [e^{2at}(2at-1)+1]/(4a^2)$; their continuous limits at $a = 0$ are t and $t^2/2$. A pilot coefficient substitutes into this expression before a compact cutoff is applied. The explicit multivariate fundamental-matrix expression is an analytical identity, not a guarantee of stable numerical inversion of $Z(t)$; strongly contracting dynamics can make that inversion ill-conditioned.

4. GLS and the order of limits

For tests ϕ_k, put $\lambda_k = A^*\phi_k$, $D_{kr} = \langle\lambda_k, S_r\rangle$ and $\Sigma_{k\ell} = \langle\lambda_k, C\lambda_\ell\rangle$. Remove exact linear dependencies. If D has full column rank, covariance-weighted weak regression has local influence and covariance

$$P = (D^T\Sigma^{-1}D)^{-1}D^T\Sigma^{-1}, \qquad V_K = (D^T\Sigma^{-1}D)^{-1}.$$

WENDy's covariance iteration motivates this construction [2], but the released iid covariance is not the correct physical-OU covariance. Under polynomial features Proposition 3 supplies the complete coupled residual weights. Its order-one term determines the small-noise efficient influence; higher

orders correct residual uncertainty at finite amplitude without by themselves removing coefficient-estimator bias.

Corollary 1 (small-noise, dense-span efficiency). Let the noise be ϵZ_C with known covariance shape C. Assume a smooth identifiable parameterization, a generating parameter interior to a compact parameter set, consistent pilots, consistent covariance-shape weights and quadrature error $o(\epsilon)$. For each fixed finite test family assume a regular local weak solution with the GLS linearization. For the covariance conclusion additionally assume uniform integrability of $\|\epsilon^{-1}(\hat{\theta}_{K,\epsilon} - \theta)\|^2$. A sufficient implementation uses a uniformly stable smooth local solve, Gaussian moment control on the local Taylor remainder, and bounded fallback outside the regular region whose probability is $o(\epsilon^2)$. Then

$$\epsilon^{-1}(\hat{\theta}_{K,\epsilon} - \theta) \Rightarrow N(0, V_K), \qquad \epsilon^{-2}\,\mathrm{Cov}(\hat{\theta}_{K,\epsilon}) \to V_K.$$

If nested test-influence spans are dense in $\mathcal{M}$, then

$$V_K \longrightarrow \mathcal{I}_{e,C}^{-1}.$$

Proof. For fixed K, a first-order expansion of the weak equations gives $D(\hat{\theta} - \theta) = \epsilon r_1 + o_p(\epsilon)$ with the GLS map above. Consistent pilots and weights change the leading map by $o_p(1)$; derivatives of pilot-dependent tests multiply the zero clean residual. Gaussianity gives the distributional assertion; the stated risk condition justifies covariance convergence. In the Gaussian dual space, finite-family information is the Gram matrix of the orthogonal projection of $C^{-1}R$ onto its test span. Density gives projection convergence, and nested spans give increasing information; continuity of inversion at positive definite $\mathcal{I}_e$ gives covariance convergence. □

The conclusion is sequential: first small noise and controlled quadrature for fixed K, then dense spans. A diagonal joint limit requires uniform remainder, conditioning and covariance-weight control that are not supplied here. This theorem concerns regular WENDy-like weak estimation with correct covariance and supplied identifiable structure, not all implementations called WENDy, finite-noise sparse selection, or a universal plateau for misspecified iid weighting. A fixed basis cutoff must be compatible with the density assumption; discarding an informative direction permanently can invalidate it.

References

[1] Dette H, Pepelyshev A, Zhigljavsky A. Optimal designs in regression with correlated errors. *The Annals of Statistics* 44 (2016) 113–152. https://doi.org/10.1214/15-AOS1361.

[2] Bortz DM, Messenger DA, Dukic V. Direct estimation of parameters in ODE models using WENDy: Weak-form estimation of nonlinear dynamics. *Bulletin of Mathematical Biology* 85 (2023) 110. https://doi.org/10.1007/s11538-023-01208-6.